Author2vec: An Averaged-Jacobian Workspace Lens for Open Embedding Encoders

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Abstract

Anthropic's *"Verbalizable Representations Form a Global Workspace in Language Models"* introduces the averaged-Jacobian lens (J-lens) and reports that a large closed decoder (Claude Sonnet 4.5, 127 layers) maintains a privileged "workspace" of verbalizable, causally-active concepts, marked by structural depth signatures of the layer Jacobian. We ask whether that apparatus survives transplant to a setting it was not built for: small open embedding encoders whose output is a single masked-mean-pooled vector. Three ingredients carry it across: a δ-broadcast reduction that collapses the encoder's position×position Jacobian to one matrix per layer (proven equal to the mean per-position Jacobian); an embedding-space projection for the L2-normalized output (proven exact); and a style lens that reads the layer Jacobian onto interpretable axes rather than the token vocabulary. We re-point the lens at a target the original could not check from the outside, authorship identity, across a 6-layer prose encoder (MiniLM), a 12-layer code encoder (JinaBERT), and a 12-layer open decoder (GPT-2). The apparatus runs on all three and yields stable, jackknife-bounded depth signatures, but the workspace *geometry* does not carry over: the averaged Jacobian shows a distinct mid-network spectral regime in the code encoder (a low-rank dip) and in GPT-2 (an inverted-U), yet these have opposite sign, and the prose encoder shows only a monotone trend. A randomly-initialized control attributes that monotone rank-with-depth trend to architecture, and the mid-network readout persistence and the decoder's next-token rank collapse to training. Identity is linearly decodable at every layer; measured against a plain-activation probe, the Jacobian readout ties it on prose and modestly exceeds it mid-network on code (best 50.8% vs. 48.8%, chance 6.7%), so the Jacobian's contribution is the geometry, not the decode accuracy. As a construct-validity check the same embeddings recover self-reported Big Five personality weakly but reliably (Openness +6.8 points; all five traits $p < 0.001$ against a shuffled-label null), while an astrological-sign negative control recovers nothing ($p = 0.96$), which is also why we make no cognitive-capacity claim. Everything runs on open models and reproduces on a laptop; every cited number is machine-extracted into an audit ledger. We make no "IQ" and no consciousness claim.

Contributions

1. **An encoder adaptation of the averaged-Jacobian J-lens** via the δ-broadcast reduction (§4.2), redefining the method for a masked-mean-pooled, non-generative encoder, with a proof that it equals the mean per-position output Jacobian (Prop. 1).
2. **An embedding-space Jacobian projection** for L2-normalized outputs (§4.3), proven to be the exact Jacobian of the normalized embedding and orthogonal to it (Prop. 2).
3. **The style lens** (§4.5): a Jacobian readout onto interpretable identity axes rather than the token vocabulary — the readout available when a pooled encoder has no clean unembedding.
4. **A jackknife-quantified reproduction study of the workspace's structural signatures on open models** (§5.3, §5.4): the averaged Jacobian yields stable depth-varying spectral signatures on

both encoders and a decoder, but the mid-network regime differs in sign across models and readouts (a low-rank dip on the code encoder, an inverted-U on GPT-2, a monotone trend on the prose encoder). The workspace geometry therefore reproduces only partially, with confidence intervals throughout, and a randomly-initialized control (three seeds, §5.3–§5.4) attributes the monotone rank trend to architecture and the readout-persistence and rank-collapse signatures to training.

5. **A depth study of authorship identity as a checkable target**: identity is linearly decodable at every layer and benchmarked against a plain probe (§5.1), commits to a single individual with depth (ignition, §5.2), and is a sign-controllable causal lever (§5.5).
6. **A measured-population construct-validity check with a negative control** (§5.6): on 2,467 psychometrically-labelled essays the same embeddings recover self-reported Big Five personality above a shuffled-label null on all five traits (Openness +6.8; all five $p < 0.001$ by a 1000-permutation test), while an astrological-sign negative control recovers nothing ($p = 0.96$).
7. **An open, reproducible, browser-native reimplementation** across prose, code, and a decoder, with a faithfulness gate and a full audit ledger.

The new method is narrow: the δ-broadcast reduction (1) and the embedding-space projection (2) are what make the averaged-Jacobian lens run on a pooled encoder at all, with the style lens (3) as the readout they enable. Contribution 4 is a reproduction study of the structural signatures, with a mixed result; 5 applies established techniques (difference-of-means directions, probing across depth, the ignition design) to a new target, authorship identity read through the layer Jacobian; 6–7 are the construct-validity check and the open reimplementation.

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1. Introduction

Recent interpretability work argues that large language models maintain a privileged, verbalizable "workspace" of representations: a small subset of activations that are reportable, controllable, and causally responsible for reasoning (Anthropic 2026). That evidence comes from a generative decoder with hundreds of billions of parameters and privileged internal access, read out with an averaged-Jacobian lens onto the token vocabulary. It is expensive to reproduce and impossible to inspect from the outside.

We ask a smaller, checkable question with the same apparatus: where, inside a network, does *who wrote this* become decidable? We move the averaged-Jacobian lens from a frontier decoder to two small, open embedding encoders, a 6-layer prose model (MiniLM) and a 12-layer code model (JinaBERT), and re-point it from reasoning onto personal writing-style identity. An encoder is a minimal testbed for the reference paper's mechanistic claims because it strips away generation: there is no autoregressive loop to smuggle information through, only a fixed map from text to a single pooled vector. It cannot speak to the paper's behavioral claims (verbal report, reasoning swaps), which need a capable generative model and are out of scope here; what it can test is whether the structural *geometry* the paper attributes to the workspace survives, which we then cross-check on an open decoder (GPT-2, §5.4).

Carrying the lens across that gap takes three ingredients (§4): a δ-broadcast reduction that collapses the encoder's intractable position×position Jacobian to one matrix per layer; an embedding-space projection that removes the meaningless radial direction of a normalized output; and a style lens that reads the layer Jacobian onto interpretable identity axes rather than the token vocabulary, where an encoder has no clean unembedding.

With these we find that author identity is linearly decodable at every layer, not only at the output (§5.1), and that an ambiguous two-author input drives the internal representation to commit to a single author increasingly with depth, above two null controls (§5.2). We then read the same averaged Jacobian's structural depth signatures on both encoders and a decoder (§5.3, §5.4) to ask whether the reference paper's "workspace" geometry reproduces on open models. It reproduces only in part: each model carries depth-dependent spectral structure with a distinct mid-network regime, but its sign differs across models and readouts (a low-rank dip on the code encoder, an inverted-U on GPT-2, a monotone trend on the prose encoder), quantified with jackknife confidence intervals. We mark replication versus new results throughout (§7); every result number is machine-extracted into the audit ledger (Appendix D) and gated by the review harness.

2. Related work

Lenses on the residual stream. The logit lens (nostalgebraist 2020) reads intermediate activations through the unembedding; the tuned lens (Belrose et al. 2023) learns an affine correction per layer. The averaged Jacobian of (Anthropic 2026) generalizes this by linearizing the *whole* map from a layer to the output and averaging over a corpus. We inherit that construction and adapt it to a pooled encoder (§4.2); our vocab lens is a logit lens over tied WordPiece embeddings, and our *style lens* (§4.5) replaces the vocabulary readout with one onto interpretable directions, the piece that is new here.

Steering and concept directions. Activation addition (Turner et al. 2023) and contrastive activation addition (Rimsky et al. 2024) steer generation by adding a direction to the residual stream; representation engineering (Zou et al. 2023) extracts such directions, often as a difference of class means. A single such direction can be a causal lever (as in refusal, mediated by one direction (Arditi et al. 2024)), and causal-mediation analysis localizes where a direction acts (Vig et al. 2020). Our identity axes are difference-of-means (Fisher) directions, used both to *read* (§4.5) and to *steer* (§5.5); the novelty is not the technique but the target, personal authorship identity, and reading it *through the layer Jacobian*.

Probing and representational geometry. Linear probes (Alain and Bengio 2017; Hewitt and Manning 2019) test what a layer linearly encodes; because above-chance probing need not imply the model *uses* the information (Hewitt and Liang 2019), we pair every decode with control-direction and shuffled-label nulls. We use leave-one-out nearest-centroid decoding as a probe of *identity* across depth (§5.1). Linear CKA (Kornblith et al. 2019) compares representations between layers, which we apply to J-lens readout geometry (§4.7). Sparse dictionary learning / SAEs (Bricken et al. 2023) decompose activations into interpretable atoms; the paper's J-space decomposition is a supervised cousin we reuse (§4.6).

Global workspace and authorship. The framing of (Anthropic 2026) draws on global workspace theory (Baars 1988) and the "ignition" of conscious access (Dehaene and Changeux 2011); we borrow the *ignition* experimental design (§5.2) but point it at identity, and we make no claim about consciousness. Our substrate is Sentence-BERT-style embeddings (Reimers and Gurevych 2019), and the downstream question, attributing text to its author from style, is classical authorship attribution / stylometry (Mosteller and Wallace 1964; Stamatatos 2009), now often approached with learned authorship embeddings (Rivera-Soto et al. 2021), here recast as a question about *where in a network* identity is computed. A known hazard is the content–style confound: apparent authorship signal can be topic signal (Wegmann et al. 2022), which we control on the model side (Appendix E) and flag as a limitation on the prose side (§7). Throughout, the J-lens, J-space, and structural signatures are the paper's; our contribution is their transplant to open encoders, the style-lens readout, the identity target, and full reproducibility.

3. Background: the averaged-Jacobian lens

We build directly on the apparatus of (Anthropic 2026), which we summarize here so that our adaptation (§4) and its boundaries are unambiguous. Quantitative figures in this section (layer counts, workspace band, J-space fractions) are that paper's *reported* values; it is a single online article, so we attribute them to it as a whole rather than by page.

The J-lens. For a causal decoder, the *averaged Jacobian* at layer ℓ is

$$J_\ell \;=\; \mathbb{E}_{t,\,t'\geq t,\,\mathrm{prompt}}\left[\frac{\partial h_{\mathrm{final},t'}}{\partial h_{\ell,t}}\right],$$

the linearized effect of a layer-ℓ activation on the final residual stream, averaged over token positions and a corpus of prompts. Read through the unembedding it yields, for any activation, a ranked list of vocabulary tokens the model is "disposed to say", a corrected logit lens that accounts for representational change across layers.

J-space and its privilege. Activations are decomposed into sparse non-negative combinations of k ($\approx$ 10–25) J-lens vectors; the *J-space component* is the part of an activation lying in that cone. Though it accounts for only $\sim$ 6–10% of a concept vector's variance, it is the part causally responsible for the concept's availability to verbal report, and it is subject to top-down control (instructing the model to "focus on X" loads X), mediates unverbalized reasoning intermediates (swapping a J-lens vector redirects downstream answers), and is required for deliberate (but not automatic) tasks.

Structural depth signatures. Four quantities, read off J_ℓ across depth (the paper's Figure 28), identify a workspace band: next-token-prediction accuracy of the readout, its *excess kurtosis* (peakiness), the *autocorrelation* of the top lens token across positions (persistence of abstract content), and the readout's *effective linear dimensionality*. All four mark a consistent sensory $\to$ workspace $\to$ motor tripartition (noisy early layers, a coherent middle band, and output-aligned late layers), corroborated by an ambiguous-input "ignition" experiment in which the representation snaps to one interpretation at workspace onset. The study is conducted on Claude Sonnet 4.5 (127 layers; workspace $\approx$ L38–92) and corroborated on Haiku/Opus.

The gap we step into. Every one of these constructs is defined for a *generative decoder* with per-position next-token logits. An embedding encoder has neither: its output is a single pooled vector. §4 is what it takes to carry the mechanistic half of this apparatus across that gap; §5.5 (causal steering) and §5.6 (a measured population) are the closest we come to the behavioral half the encoder cannot reach.

4. Method

Setup. We study two open embedding encoders: `sentence-transformers/all-MiniLM-L6-v2` (Wang et al. 2020) (prose; 6 layers, $d = 384$, vocab 30,522) and `jinaai/jina-embeddings-v2-base-code` (Günther et al. 2023) (code; 12 layers, $d = 768$, vocab 61,056). Each maps a passage to a masked-mean-pooled vector $p = \frac{1}{T}\sum_t h_{L,t}$; the shipped embedding is its L2-normalization $\hat{p} = p/\|p\|$. A faithfulness gate (`bin/spike`) asserts our native candle forward reproduces the shipped fastembed embeddings at cosine > 0.99 before any Jacobian is trusted.

4.1 The problem an encoder poses

The causal J-lens is defined per token position toward next-token logits. An encoder emits a single pooled vector and no logits, so the per-position causal Jacobian is undefined here.

4.2 δ-broadcast averaged Jacobian

We perturb every source position by a shared displacement δ and differentiate the pooled vector, collapsing the position$\times$position Jacobian to one matrix per layer:

$$J_\ell \;=\; \frac{1}{T}\,\frac{\partial p}{\partial \delta}\Big|_{\delta=0} \;=\; \frac{1}{T}\sum_{t=1}^{T}\frac{\partial p}{\partial h_{\ell,t}} \;\in\; \mathbb{R}^{d\times d},$$

which is exactly the mean over source positions of the per-position output Jacobian (Prop. 1, Appendix A), independent of T. It is built by batched central finite differences ($\varepsilon = 0.05$, pure gemm), not per-dimension autograd. `lib.rs:layer_jacobian`.

4.3 Embedding-space projection

The shipped embedding is the L2-normalized $\hat{p} = p/\|p\|$, so we compose J_ℓ with the Jacobian of the normalization map to obtain the Jacobian of the actual output:

$$J_{\text{emb}} \;=\; \tfrac{1}{\|p\|}\,(I - ee^\top)\,J_\ell \;=\; \frac{\partial \hat{p}}{\partial \delta}, \qquad e = \hat{p}.$$

This is exact, not an approximation, and satisfies $e^\top J_{\text{emb}} = 0$ (Prop. 2, Appendix A): it discards precisely the radial component that renormalization annihilates. `lib.rs:to_embedding_jacobian`.

4.4 Vocabulary (logit) lens

Standardized $J \cdot h$ times the tied WordPiece embeddings $W_U \to$ top tokens. Noisy here (the checkpoint ships no trained MLM head); reported but not load-bearing. `lib.rs:vocab_topk`.

4.5 Style lens

We read the Jacobian onto interpretable identity axes instead of the vocabulary:

$$s_a(h) \;=\; A_a \cdot \mathrm{normalize}(J_{\mathrm{emb}}\, h), \qquad A_a = \mathrm{normalize}\,(\bar{c}_a^{+} - \bar{c}_a^{-}),$$

each axis a unit difference-of-class-means (Fisher) direction over the reference embeddings. Shipped axes, prose: `male` $\leftrightarrow$ `female`, `educated=England` $\leftrightarrow$ `rest`, `educated=US` $\leftrightarrow$ `rest`, `raised=England` $\leftrightarrow$ `rest`, `raised=US` $\leftrightarrow$ `rest`; code: `Systems` $\leftrightarrow$ `rest`, `Scripting` $\leftrightarrow$ `rest`. `lib.rs:style_scores`, `axis`, `author_axes`.

4.6 J-space decomposition

Non-negative matching pursuit over unit rows of $W_U J_\ell$ yields a sparse concept set and the fraction of activation variance it captures. `lib.rs:jspace_nmp`.

4.7 Structural depth signatures

Per layer: stable rank $\|J\|_F^2/\sigma_1^2$, effective dimension (participation ratio $\|J\|_F^4/\|J^\top J\|_F^2$), verbalizability (excess kurtosis of the vocab-lens readout), layer×layer linear CKA, and, completing the paper's four-metric set, autocorrelation (lag-1 readout persistence vs. a position-shuffled null). `lib.rs:stable_rank`, `effective_dim`, `excess_kurtosis`, `linear_cka`, `readout_autocorrelation`.

5. Experiments and results

5.1 Identity is linearly decodable at every layer

Decoding author identity by leave-one-out nearest-centroid on the internal per-layer Jacobian readout beats chance at every layer. For prose, the per-layer rates $[5.2, 7.2, 8.8, 8.0, 7.2, 8.8]$% all clear the 1.8% chance (output ceiling 14.4%); even the weakest (5.2%, Wilson $[3.1, 8.7]$, n=250) excludes chance (binomial $p <$ 0.001). For code the rates run far higher, up to 50.8% against a 6.7% chance. The best internal code layer (50.8%, $[44.6, 56.9]$) matches the output-embedding ceiling (46.8%, $[40.7, 53.0]$): the intervals overlap by ~ 8 points, so the internal readout is *as good as* the output, not better. Identity is linearly present inside the layers, not only at the output. *(Wilson 95% CIs, decode sample n=250; per-layer values in ledger `identity_*`, bounds in `ci_identity_*`.)*

Benchmark against a plain probe. The averaged Jacobian is more expensive than the obvious baseline: decode identity from the raw mean-pooled activation at each layer, with no Jacobian at all. We run both. On prose they are statistically indistinguishable, best layer 8.8% (J-lens) versus 8.4% (plain probe) and mean 7.5% versus 7.0% across layers, well inside the Wilson intervals above. On code the Jacobian readout does modestly better, mean 44.6% versus 40.9% and best layer 50.8% versus 48.8%, with the gap concentrated in the middle of the network (e.g. layer 8, 42.0% versus 31.6%); at n=250 each per-layer gap is only two to three standard errors, so we read the mid-network edge as suggestive. The Jacobian readout never underperforms a plain probe and modestly outperforms it mid-depth on code, but decode accuracy is not where the averaged Jacobian earns its keep. That is the structural geometry (§5.3, §5.4) and the steerable direction (§5.5); §5.1 establishes only that identity is linearly present at every depth, robustly to the readout.

5.2 Identity ignition: commitment sharpens with depth

We adapt the paper's ambiguous-input ignition to identity. For an author pair (A, B) we build a per-depth difference-of-means axis $\hat{u}_\ell = \widehat{c_{A,\ell} - c_{B,\ell}}$ from their *training* passages, then blend two *held-out* passages at the

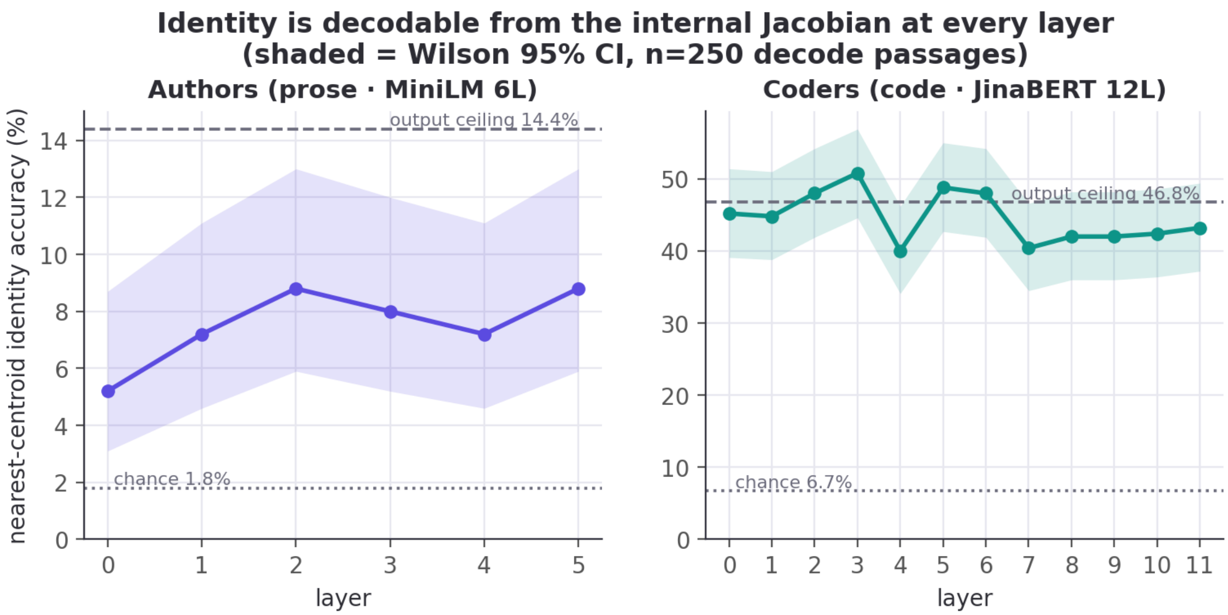

Figure 1: Identity accuracy decoded from the internal Jacobian at each layer, vs. chance (dotted) and the output-embedding ceiling (dashed).

input embedding, $h_0(\alpha) = (1-\alpha)h_0^B + \alpha h_0^A$, sweep $\alpha \in [0, 1]$, and read the commitment $s_\ell(\alpha) = \hat{u}_\ell(\bar{p}_\ell(\alpha) - m_\ell)$ at every depth (15 pairs). A graded layer ramps linearly in α; an *ignited* layer snaps.

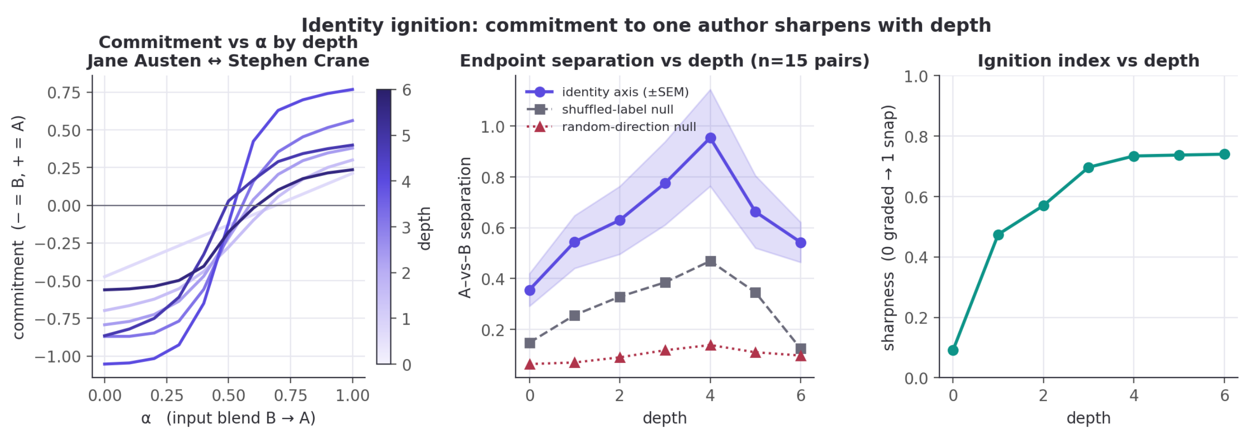

Figure 2: Identity ignition (MiniLM). Left: commitment vs. α by depth for one pair (input-linear at depth 0, stepped by mid-depth). Middle: A-vs-B separation rises to a mid-network peak, far above the random null (~7×) and clearly above the shuffled null. Right: the ignition index (transition sharpness) rises with depth.

The representation commits to one author, increasingly with depth. Endpoint separation along the identity axis rises from 0.36 at the input to a mid-network peak of 0.95 at depth 4 (of 6; ± 0.19 SEM over 15 author pairs), then eases to 0.54 at the output. It dominates both controls at every depth: the random-direction null sits at 0.06–0.14 (a $\sim 7\times$ margin at the peak) and the shuffled-label null at 0.12–0.47. The effect is therefore identity-specific, not a generic consequence of blending inputs. The ignition index (transition sharpness, 0 = graded, 1 = all-or-none) climbs from 0.09 at the input (where the readout is, correctly, linear in the blended input) to 0.73–0.74 by mid-depth and holds.

The same holds for code, more strongly. On the 12-layer code encoder the ignition index climbs from

0.09 at the input to a peak of 0.82 (depth 11 of 12). Through the early-mid layers the identity axis separates developers far above the random-direction null: a $\sim 13\times$ margin at depth 4 (1.21 vs 0.09; raw separation peaks slightly earlier, 1.56 ± 0.22 at depth 3). It is a sharper version of the same effect, consistent with code identity being more linearly accessible overall (§5.1). Separation spikes higher still at the final layer (2.33), but that depth is noisy: its wide ± 0.53 SEM and a jumping null are why we feature the stable early-mid layers.

Two caveats bound this reading. It rests on a 6-layer encoder and 15 pairs. And although the separation control confirms the axis is identity-specific, we cannot fully exclude that some of the depth-wise sharpening reflects generic late-layer nonlinearity.

5.3 Structural signatures across depth

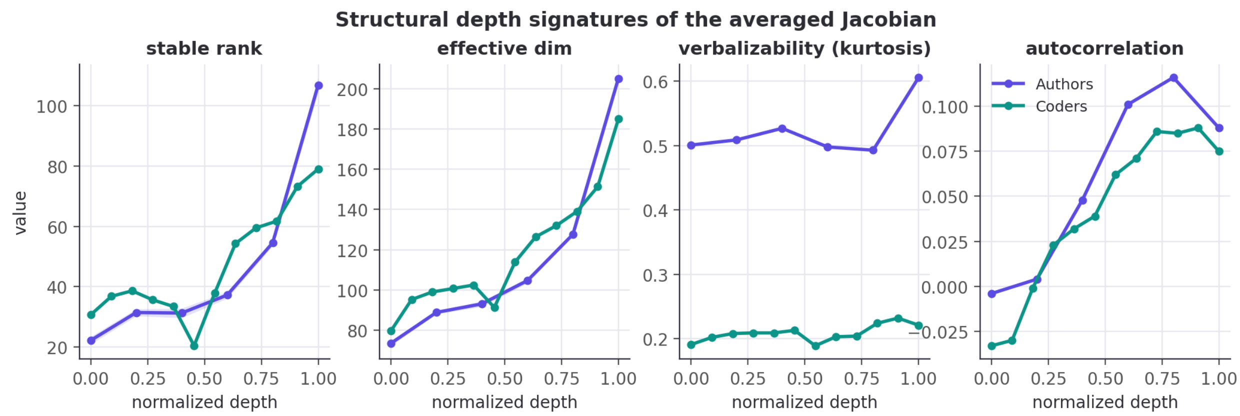

Figure 3: Structural depth signatures of the averaged Jacobian (normalized depth). Shaded: delete-a-group jackknife 95% CI (MiniLM; the code encoder's per-passage jackknife is compute-deferred, §7).

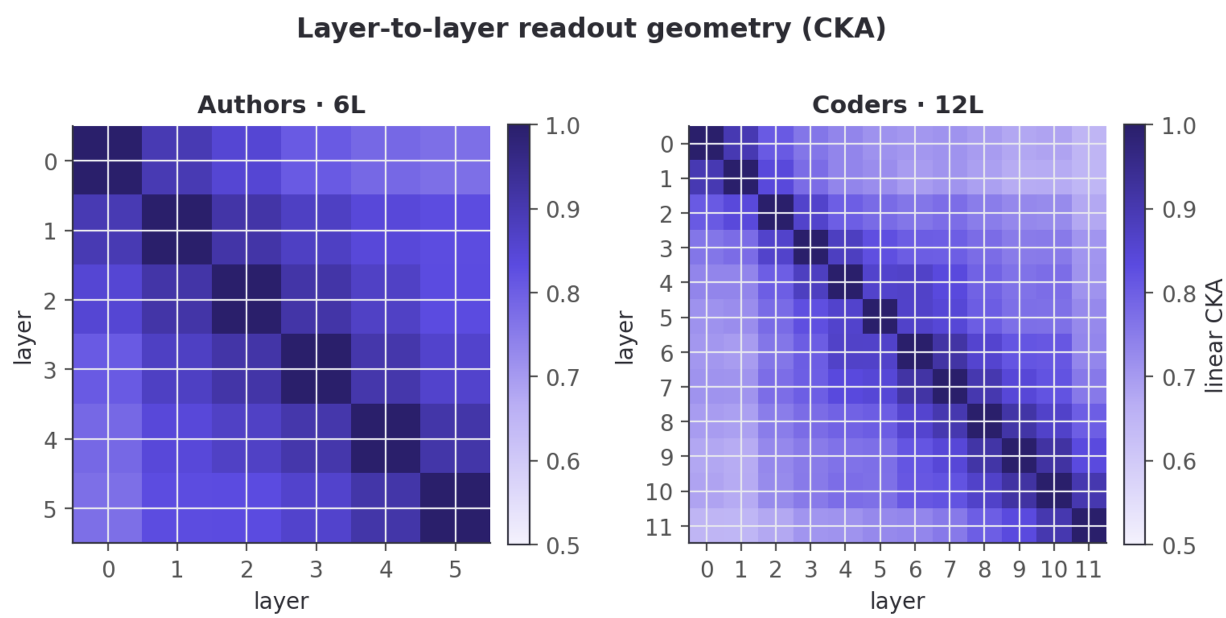

Figure 4: Layer-to-layer readout geometry (linear CKA).

We read four structural signatures off the averaged Jacobian at each depth, with delete-a-group jackknife

95% confidence bands over passage folds on the two spectral signatures (Fig 2 for MiniLM, Fig 7 for GPT-2; the 12-layer code encoder's per-passage Jacobian is expensive, so we report its point estimates and defer its per-layer band, §7). Stable rank and effective dimension are two views of the same singular spectrum and move together; verbalizability is nearly flat on both encoders (prose ≈ 0.5, code ≈ 0.2) and carries little depth signal, so the spectral profile is the informative part.

The two encoders do not share one depth profile. On prose (MiniLM) stable rank rises monotonically with depth (22.1 at layer 0 to 106.9), lowest at the very first layer, with no mid-network dip. On code (JinaBERT) it instead has a local minimum mid-network (20.3 at layer 5 of 12), a low-rank dip flanked by higher-rank early and late layers, with a matching dip in effective dimension; effective dimension otherwise rises with depth in both (prose $73 \to 205$; code $80 \to 185$). The mid-network low-rank bottleneck appears on the code encoder but not the prose one, so it is a property of the code model, not the reference paper's workspace band. On MiniLM the rise is many times its jackknife band (Fig 2), so the profile is well-determined; the same band on the code encoder is what we defer.

The fourth signature, autocorrelation (persistence of the readout across positions), is the one direction that agrees across models: near zero or negative at the shallowest layers and rising through the middle in both (prose peaks at 0.12 around layer 4; code climbs to ≈ 0.09 by layers 8–9), a readout-persistence trend consistent with (Anthropic 2026). We do not see the paper's sensory→workspace→motor tripartition; at 6–12 layers there is too little depth for it.

Untrained control: which of these is training-induced? To separate learned structure from architecture we rerun the pipeline on a randomly-initialized MiniLM (HuggingFace `_init_weights`; three seeds, `randinit.rs`), reading the same signatures off the untrained network (Fig 12). The monotone stable-rank rise *survives at init*: 109.2 at layer 0 to 215.4 at layer 5 (across-seed SD ≤ 3.5), the trained encoder's increasing shape but 2–5× higher. The rise with depth is thus architectural; training *compresses* the spectrum, most at the shallow layers (trained layer 0 is 22.1, a 5× drop). Verbalizability is likewise unchanged at init (0.46–0.54). Autocorrelation is the exception: flat at ≈ 0 (max $|\cdot| = 0.005$) at every depth untrained, against the trained peak of 0.116. The mid-network readout-persistence trend — the one signature that agreed across models — is therefore training-induced; the bare rank-rise is not. (The 12-layer code encoder's untrained control, like its per-layer jackknife band, is deferred, §7.)

5.4 Structural signatures on a decoder (GPT-2)

Everything above is on *encoders*. If the depth-wise workspace geometry is a real property of the J-lens apparatus and not an artifact of masked-mean pooling, then reading a generative decoder with the *same* averaged-Jacobian machinery should reproduce the paper's Figure-28 depth signatures. Hypothesis: on an open decoder we will see (i) the four Jacobian signatures vary with depth, and (ii) a decoder-only signature, next-token logit-lens accuracy, rise sharply in the late layers, marking the "motor" regime where representations turn toward the output. We do not expect a sensory→workspace→motor tripartition: GPT-2 is 12 layers, far short of the paper's ~100.

Method. On GPT-2 (Radford et al. 2019) (`openai-community/gpt2`; 12 layers, d=768) we run the δ-broadcast averaged Jacobian, made causal (perturb every position, read the *last* position, the next-token driver), over prose prompts. The four signatures use the *same* `lib.rs` functions as the encoders, and verbalizability is read through GPT-2's real tied unembedding (no approximation). This is the paper's mechanistic apparatus on an open model that generates.

Result. Next-token accuracy is near zero through the first six layers, then climbs steadily to 22.6% at the output: the late layers are the motor regime (hypothesis ii). The two spectral signatures trace an inverted-U — stable rank low at the input (2.6), high through the middle (6.7 at layer 5), collapsing to near rank-one at the output (2.0); effective dimension likewise ($5.8 \to 33.5 \to 4.1$) — with jackknife 95% bands in Fig 7, where the mid-network peak separates from the low-rank output end but not from the input. Autocorrelation rises with depth to a peak of 0.16 (layer 10) before easing to 0.11 at the output.

This is a different shape from the encoders, not a sharpened version of them. The decoder's mid-network is a stable-rank *maximum*; the code encoder's (§5.3) is a stable-rank *minimum*. The two profiles

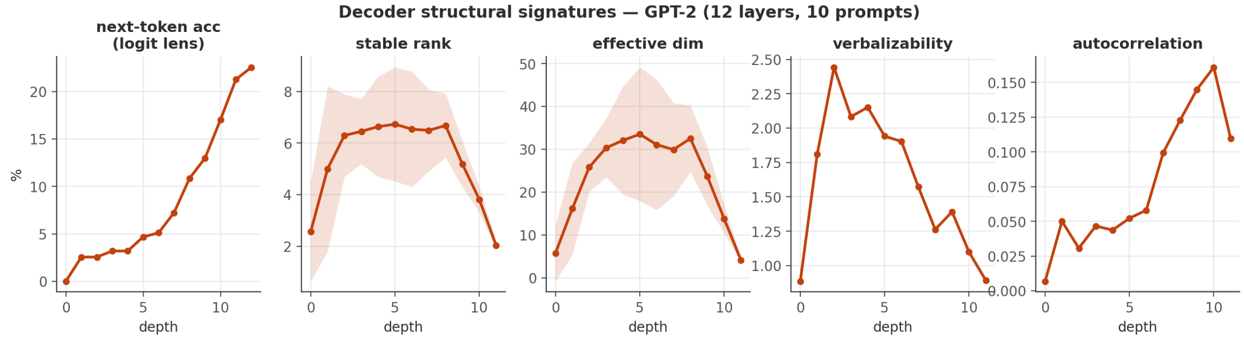

Figure 5: Decoder structural signatures on GPT-2 (Figure-28 series + next-token accuracy).

have opposite curvature, and they are read under different reductions. The encoders average the Jacobian over all pooled positions, while the decoder reads the last position (the next-token driver), which necessarily collapses to low rank as the network commits to a single output token. So the decoder's low-rank output end is partly manufactured by the readout, not a "motor" property in the paper's sense (there $J_\ell \to$ identity, full rank). We therefore do not claim the encoder and decoder exhibit the *same* workspace geometry. What both show is that the averaged Jacobian carries depth-dependent spectral structure with a mid-network regime distinct from the endpoints; the *sign* of that regime depends on the model and the readout. The shapes matter more than the levels here: final-layer next-token accuracy (22.6%) is low, given archaic literary prose, a 48-token context, and 10 prompts.

Untrained control sharpens the decoder reading. The same random-init test on GPT-2 (three seeds; Fig 12) settles the caveat above. At init the stable rank rises monotonically to 345.8 at the output — no inverted-U, and no collapse. The trained decoder's low-rank output end (2.0) is thus training-induced, not a readout artifact: the identical last-position readout on random weights gives the *highest* rank at the output, because an untrained network commits to no next token. Consistently, untrained next-token accuracy is 0.0 at every depth (exact chance) against the trained rise to 22.6% — the motor regime is entirely learned. Autocorrelation carries a modest architectural baseline (≈ 0.07) that training lifts into the depth-rising trend peaking at 0.161. So the inverted-U's output arm and the next-token rise are training signatures; the underlying rank-grows-with-depth is not.

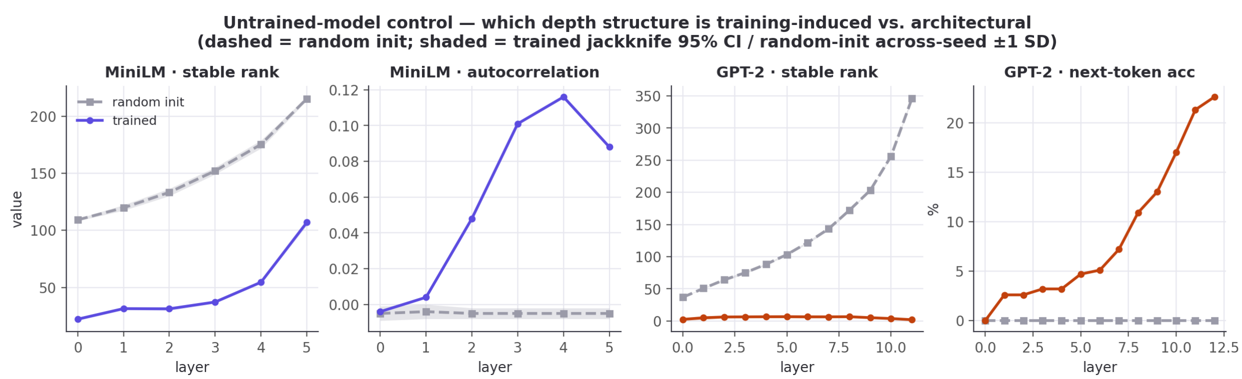

Figure 6: Untrained-model control (§5.3, §5.4). The identical structural pipeline on randomly-initialized MiniLM and GPT-2 (three seeds each). The monotone stable-rank rise is present at random init (architectural); the mid-network autocorrelation trend, the decoder's output-end rank collapse, and next-token accuracy are absent at init (training-induced). Dashed = random init; shaded = trained jackknife 95% CI / random-init across-seed ±1 SD.

5.5 The identity direction is a causal lever

Reading a direction is weaker than showing the network acts on it. We form the residual steering direction $\hat{\delta} = \widehat{J_{\text{emb}}^{\top} A}$ from a style axis A and inject $\alpha\hat{\delta}$ (broadcast to every position) at the mid layer. This drives the output's loading on A through a large, sign-controllable swing: monotonic from negative (at $\alpha = -6$) through zero (unperturbed) to positive (at $\alpha = +6$), saturating (and for gender slightly reversing) at the extremes. The effect has the same sign across all five prose identity axes (gender, education, upbringing), though not the same magnitude: the gender axis swings least (0.80), the other four up to 1.11. A matched-norm random direction, injected identically, leaves the loading flat (total drift ≤ 0.16 over the same sweep). The mid layer therefore holds the identity direction as something the rest of the network *acts on*, not merely correlates with.

A leakage-free check on the decoder. The encoder steer reads back on the same axis it perturbs, so we repeat the test on GPT-2 with an independent readout. We build a concept-context direction as the difference of mid-layer mean residuals between concept-primed and neutral prompts (not the target's unembedding), inject $\pm\alpha\hat{\delta}$ at the mid layer, and measure held-out concept-token log-probability. The direction is a sign-controllable handle, raising concept-token log-prob by +0.10/+0.22/+0.44 (money / music / war) for $+\alpha$ and lowering it below baseline for $-\alpha$; the mean effect is +0.254 versus −0.085 for a matched-norm random control. It is real and bidirectional but small (it shifts the distribution, not the argmax output) and rests on 3 concepts × 6 prompts, so we report it as the mechanistic prerequisite the paper's behavioral experiments rely on, not the behavioral result itself. Redirecting a reasoning intermediate to change a final answer needs a model that can reason; that is future work (§7).

5.6 Measured constructs, a negative control, and why we make no "IQ" claim

Every attribute so far (gender, region, systems-vs-scripting) is a *label we assigned.* The sharpest test of whether these embeddings carry real psychological signal is to hand them a population whose attributes were measured by someone else, with a validated instrument, and ask whether the signal survives. Hypothesis: if prose style encodes personality at all, an embedding built with no knowledge of psychology should recover self-reported Big Five traits above chance. Method: the Pennebaker & King stream-of-consciousness corpus (Pennebaker and King 1999), 2,467 essays, each labelled with the writer's Big Five traits from a self-assessment questionnaire (ground truth, not our guess), embedded with the *same* MiniLM used throughout. For each trait we run leave-one-out nearest-centroid classification (high vs. low group) against the majority-class baseline, with a shuffled-label null.

Result: every trait beats a shuffled-label null, and the null is flat. Openness is strongest at 58.3% (majority 51.5%; +6.8 points), then Neuroticism 56.3% (+6.3), Conscientiousness 55.1% (+4.3), Extraversion 55.4% (+3.6), Agreeableness 55.3% (+2.2), a mean lift of +4.6 points over the majority baseline. A 1000-permutation test, which shuffles the trait labels and re-runs the whole classifier, puts all five traits at $p < 0.001$ against that null (which sits at chance, 49.9–50.1%). The permutation test is against label-shuffling, not the majority baseline, so we also check each trait's Wilson 95% interval against its majority class: Openness clears it comfortably (58.3%, $[56.3, 60.2]$ vs. 51.5%), but Agreeableness only marginally (55.3%, $[53.3, 57.2]$ vs. 53.1%), its lower bound barely exceeding the baseline. The lift over majority is weak for the lowest traits, as expected. This is the *weak-but-real* ceiling the personality-from-text literature reports (Mairesse et al. 2007), and Openness-leads-the-pack is a standard finding at social-media scale (Schwartz et al. 2013). It is a positive result on an externally-measured population: MiniLM, trained for no psychological task, recovers self-reported traits a few points above chance.

A negative control: what the method does *not* recover. A positive result could still be spurious structure, so we pair it with an attribute that *should* be unrecoverable and check that the method stays silent. The Blog Authorship Corpus (Schler et al. 2006) labels every blogger with gender, age, and astrological sign: the first two have linguistic correlates, the third has none. We pool posts to the author level (138 authors with a known sign; each author is the mean of their post embeddings) and run the *identical* leave-one-author-out nearest-centroid classifier on all three, each against a shuffled-label null.

Age band recovers strongly and significantly: 53.6% (majority 43.5%) against a 31.6% shuffled-label null,

Figure 7: Big Five recovery from prose: leave-one-out accuracy vs. majority baseline, with the shuffled-label null flat at chance.

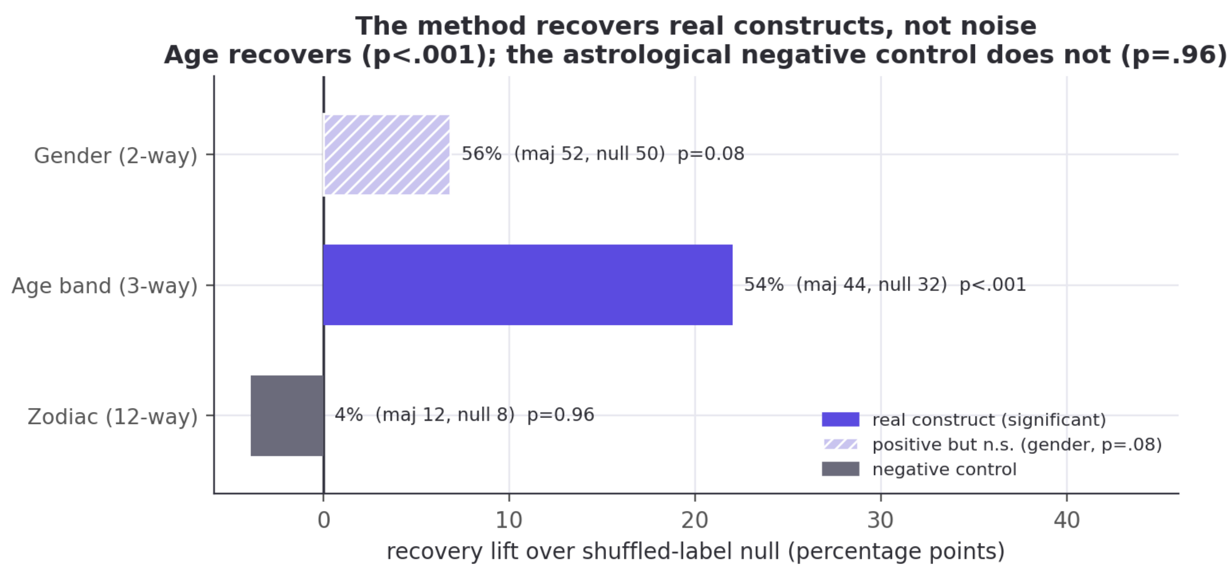

Figure 8: Construct validity on one corpus (author-level, 1000-permutation test): writer age recovers ($p < 0.001$) and the astrological-sign negative control does not ($p = 0.96$); gender is directionally positive but not significant at this sample ($p = 0.08$).

$p < 0.001$ over 1000 permutations, while astrological sign shows *nothing*: 4.3%, *below* both its majority baseline (11.6%) and its shuffled null (8.2%), $p = 0.96$ (the real labels do worse than 96% of random relabelings). Gender lands in between, directionally positive but not significant at this modest sample (56.5% against a 49.7% null, $p = 0.08$), consistent with a weak signal that 138 author-level points are underpowered to confirm. The same pipeline that recovers age (a construct with linguistic correlates) recovers nothing from astrological sign — evidence that the signal tracks real constructs, not pipeline artifacts. This check reads the pooled output embedding, not the layer Jacobian, so it validates the *embeddings and corpus*, not the lens; its role is to license, and to bound, the identity signal the rest of the paper reads through J_ℓ.

Those two results together are why we make no "IQ" claim. A validated psychometric construct tops out a few points over chance; a loaded, poorly-operationalized one like "intelligence" would fare no better and would invite far worse misreading. Whether one can steer a *capability* score (a vocabulary test administered to a decoder) is a question for a capable model with careful construct validation, and is left to future work.

5.7 Ablations

Rather than a hyperparameter sweep, the study carries several *built-in* ablations. Readout: §5.1 decodes identity from the layer Jacobian and from the raw activation with no Jacobian, and the two are within noise on prose (the Jacobian modestly ahead on code), so the identity-at-every-layer result does not depend on the Jacobian readout. Model class: the GPT-2 decoder (§5.4) runs the *same* apparatus on a generative model; the spectral signature is depth-varying there too, though its shape differs from the encoders. Training vs. architecture: §5.3 and §5.4 rerun the structural pipeline on randomly-initialized checkpoints of MiniLM and GPT-2 (three seeds each), isolating the signatures that are training-induced (the autocorrelation trend, the decoder's rank collapse, next-token accuracy) from the one already present at init (the monotone rank rise). Controls-as-ablations: every causal claim ablates its direction against a matched-norm random and/or shuffled-label null (§5.2, §5.5). Two things we did *not* do, and flag as limitations rather than strengths (§7): the prose-vs-code contrast confounds depth with modality, architecture, and dimension, so it is not a clean depth ablation; and we did not sweep the finite-difference ε or the Jacobian context length, so the structural curves' robustness to those is untested. Bit-for-bit determinism at a fixed ε says nothing about sensitivity to it.

6. Discussion

What transfers, and what does not. Read with the same averaged-Jacobian apparatus as (Anthropic 2026), these models show *some* workspace-like structure around identity: the fingerprint is linearly present at every layer (§5.1), it becomes increasingly separable and commits to a single individual with depth (§5.2), and the identity direction is a causal lever (§5.5). The apparatus itself carries across to code and to a generative decoder, letting us ask a question the paper never did: *whose style is this?* What does *not* reproduce is the workspace's structural depth signature. The averaged Jacobian does carry a distinct mid-network spectral regime, but with opposite sign on the code encoder (a low-rank dip) and the decoder (an inverted-U), under different readouts, and none at all on the prose encoder (§5.3, §5.4). What transfers is the *apparatus* and the *linear and causal accessibility* of identity, not a reproduced "workspace band."

What it does not support. None of this is evidence of reportability, reasoning, or anything cognitive. Our decoder steering moves a distribution, not an answer (§5.5); a validated psychometric construct recovered from these embeddings tops out only a few points over chance (§5.6). The "workspace" here is a claim about representational geometry and linear accessibility, not about a model *knowing* or *reporting* who you are. Personal writing style is a low-dimensional, causally-active, depth-localized direction in these models' representations — a narrower claim than "the AI knows you," and one that holds on open models.

What the encoder isolates. Stripping away generation removes the decoder story's main confound (autoregressive leakage) and lets the accessibility claims stand or fall on geometry alone. Running the identical apparatus on a decoder (§5.4) cross-checks the *method* and, in the process, shows the structural signature to be readout-dependent — a finding about the lens, not a reproduction of the workspace.

7. Limitations and threats to validity

Replication vs. novel. The J-lens, J-space decomposition, and the four structural signatures are the paper's (Anthropic 2026); our contribution is the encoder adaptation, the style-lens readout, the identity target, and reproducibility.

Construct scope. We measure identity commitment and a measured-personality signal (§5.6), and we borrow the *ignition* experimental design without its consciousness conclusion.

The ignition sharpening has a confound. The ignition index rising with depth (§5.2) is measured on the identity axis, and the separation control confirms that axis is identity-specific (real separation runs $\sim$7–13$\times$ above a random-direction null). But we cannot fully exclude that *some* of the depth-wise sharpening is generic late-layer nonlinearity: any readout of a linearly-blended input can become more nonlinear with depth. What the controls establish is that the *axis* carries identity; the raw sharpening curve should be read as suggestive, not decisive. Our shuffled-label null is also imperfect (its two groups still contain real passages, so it sits above zero); the random-direction null is the cleaner floor.

Statistical power. Ignition uses 15 author/coder pairs and a 7- or 11-point α grid; the structural signatures average over 32–96 passages (encoders) and 10 prompts (GPT-2). These are small. For the structural signatures we report delete-a-group jackknife 95% confidence bands over passage folds on MiniLM and GPT-2 (Fig 2, Fig 7), which bound the single-run uncertainty of the passage-averaged estimate; the 12-layer code encoder's per-passage Jacobian is expensive to average, so its per-layer jackknife is deferred and we report point estimates there. For ignition we report standard deviations and the raw per-depth series. The ignition *index* at each depth averages only over the pairs whose endpoints separate along the axis (5–14 of 15, fewest at the shallowest layers).

Untrained-model control (partial). We run the structural pipeline on randomly-initialized MiniLM and GPT-2 (three seeds each; §5.3, §5.4, Fig 12), separating training-induced structure from architecture on those two models. The monotone stable-rank rise is present at init, hence architectural; the mid-network autocorrelation trend, the decoder's output-end rank collapse, and next-token accuracy are absent at init, hence training-induced. We therefore state the depth signatures with that split explicit, not as an undifferentiated "workspace geometry." Deferred is the 12-layer *code* encoder's control — the same follow-up as its per-layer jackknife band (§5.3) — so whether its mid-network low-rank dip is architectural or learned is still open.

Reproducibility caveats. Structural signatures reproduce bit-for-bit on MiniLM across runs; on the 12-layer models (JinaBERT, GPT-2) we have not verified GPU-reduction determinism, and the JinaBERT numbers differ from an earlier site build generated with different (undocumented) settings, so we report a run with documented settings (`JLENS_JAC_LEN=64`) and quote the jackknife CIs as the uncertainty on those point estimates. The single-averaged, single-direction lens is lossy; the vocab lens is noisy on encoders (no trained MLM head); finite-difference ε introduces $O(\varepsilon^2)$ error.

Scope. 55 authors / 15 coders on laptop-scale models demonstrate the *mechanism*; that a frontier model trained on someone's millions of words holds a sharp, personal fingerprint of *that individual* is a well-motivated extrapolation, not something these data settle. The behavioral half of the paper (the decoder steering, §5.5) is the hardest to carry over and is where a small open decoder's limited capability bites; we state it as directional, with negative results reported as such.

Content vs. style (prose). Our public-domain authors write about different subjects, so some prose "identity" signal is inevitably topic, not style (Wegmann et al. 2022). Unlike the code side, where we can control the task explicitly (Appendix E), we do not fully disentangle the two on the prose side; prose identity should be read as *style-or-topic* fingerprinting, not pure style.

Ethics and broader impact

This work shows that authorship identity and correlates of demographic and psychological attributes (gender, region, age, Big Five personality) are linearly recoverable from prose and code embeddings, and that these directions are causally steerable. The apparatus is dual-use: the same machinery can support de-anonymization of pseudonymous authors, including developers from name-scrubbed code (Caliskan-Islam et al. 2015; Narayanan et al. 2012), and non-consensual inference of sensitive attributes, enabling surveillance or profiling (Hovy and Spruit 2016). We take three positions. Consent and scope: our prose subjects are public-domain authors and our developer corpus is public, git-attributed, and name-scrubbed; we make no per-individual attribution claim and explicitly decline the AI-usage and "intelligence" inferences the method could invite (§5.6, Appendix E). Construct restraint: the astrological-sign negative control ($p = 0.96$) is included to show that a recovered signal must be validated against a real construct before use, and to caution against reading weak correlational demographic signals as ground truth. Mitigation: the demographic directions we find can also be *removed* (nullspace projection / INLP (Ravfogel et al. 2020)), and we regard attribute erasure and stylistic obfuscation as the appropriate defensive counterpart to this analysis. We release only open models and public corpora, and no tool aimed at identifying specific private individuals.

8. Reproducibility

Models. `sentence-transformers/all-MiniLM-L6-v2` (prose; 6 layers, d=384) and `jinaai/jina-embeddings-v2-base-code` (code; 12 layers, d=768), loaded natively via candle and gated (`bin/spike`) to reproduce the shipped fastembed embeddings at cosine > 0.99.

Pipeline.

```
cargo run -p corpus --release                  # prose corpus -> person2vec-minilm.{json,bin}
cargo run -p corpus --bin coders --release     # code corpus  -> person2vec-coders.{json,bin}
cargo run -p jlens  --bin spike     --release              # faithfulness gate
cargo run -p jlens  --bin jlens     --release -- <ds> <N>  # -> person2vec-jlens-<ds>.json
cargo run -p jlens  --bin steer     --release -- <ds>      # -> person2vec-identity-<ds>.json
cargo run -p jlens  --bin fingerprint --release -- <ds>    # -> person2vec-fingerprint-<ds>.json
 ↪  (App. E)
cargo run -p jlens  --bin ignition    --release -- <ds>    # -> person2vec-ignition-<ds>.json
 ↪  (sec 5.2)
cargo run -p jlens  --bin steer_bundle --release -- <ds>   # -> person2vec-steer-<ds>.json
 ↪  (sec 5.5)
cargo run -p jlens  --bin decoder_structural --release -- 10  # -> decoder-structural-gpt2.json
 ↪  (sec 5.4)
cargo run -p jlens  --bin decoder_steer --release          # -> decoder-steer-gpt2.json
 ↪  (sec 5.5)
./target/release/embed_texts minilm essays_in.json essays_out.json   # embed Pennebaker essays
 ↪  (sec 5.6)
python3 jlens/paper/recover_bigfive.py        # -> person2vec-bigfive.json  (Big Five recovery)
 ↪  (sec 5.6)
python3 jlens/paper/recover_blog_demographics.py  # -> person2vec-blog-demographics.json  (neg.
 ↪  control, sec 5.6)
python3 jlens/paper/build_ledger.py           # -> jlens/paper/ledger.json  (every cited number)
python3 jlens/figures/make_figures.py         # -> jlens/figures/*.png
```

`<ds>` $\in$ {`minilm`, `coders`}. Env vars: `JLENS_DEVICE` (`metal`|`cpu`), `JLENS_JAC_LEN` (Jacobian context; default 64 for the encoder binaries, 48 for `decoder_structural`), `JLENS_CHUNK` (finite-difference batch; smaller = less GPU memory, identical numbers). Determinism: the structural signatures reproduce bit-for-bit across runs on MiniLM (verified); run-to-run determinism on the 12-layer models (JinaBERT, GPT-2) is not verified, and we quote jackknife CIs as the uncertainty on those estimates (§7). The ignition null uses a fixed splitmix64 seed. The `/jlens` viewer does no linear algebra; all quantities are precomputed offline and shipped as static JSON.

Auditability. Every quantitative claim resolves through the audit ledger (Appendix D, `build_ledger.py`) to a source bundle and JSON path; every method claim resolves through `method_map.md` to a `file:line`.

Appendix A — δ-broadcast reduction and the embedding projection

A masked-mean-pooled encoder produces the pooled vector $p = \frac{1}{T}\sum_t h_{L,t}$ and the L2-normalized output $\hat{p} = p/\|p\|$. The full sensitivity of p to the layer-ℓ residuals is a rank-4 object $\partial p_i/\partial h_{\ell,t,j}$ of size $d \times (T \times d)$, intractable to form and to average across variable-length prompts. Perturbing every source position by the same $\delta \in \mathbb{R}^d$ collapses it to one $d \times d$ matrix.

Proposition 1 (the δ-broadcast Jacobian is the mean per-position Jacobian). Let $h_{\ell,t}(\delta) = h_{\ell,t} + \delta$ for all t and $J_\ell := \frac{1}{T}\,\partial p/\partial\delta\,|_{\delta=0}$. Then

$$J_\ell \;=\; \frac{1}{T}\sum_{t=1}^{T} \frac{\partial p}{\partial h_{\ell,t}}.$$

Proof. Writing $p = p(h_{\ell,1}(\delta), \dots, h_{\ell,T}(\delta))$, the chain rule gives $\partial p/\partial\delta = \sum_t (\partial p/\partial h_{\ell,t})(\partial h_{\ell,t}/\partial\delta) = \sum_t \partial p/\partial h_{\ell,t}$, since $\partial h_{\ell,t}/\partial\delta = I$. Divide by T. □

So J_ℓ is not an approximation to the per-position Jacobian but exactly its average over source positions. Each column c is estimated by central finite differences with $\pm\varepsilon e_c$ broadcast to all positions,

$$(J_\ell)_{\cdot c} \;\approx\; \frac{p(+\varepsilon e_c) - p(-\varepsilon e_c)}{2\varepsilon\, T},$$

whose truncation error is $O(\varepsilon^2)$ by Taylor expansion; columns are computed in batches (pure gemm) via one `forward_from(ℓ)` per batch. On a causal decoder (§5.4) the identical construction reads the last position $p = h_{L,\text{last}}$ (the next-token driver) rather than the mean, giving the causal analog.

Proposition 2 (the projection is the exact normalized Jacobian). With $e = \hat{p}$, $J_{\text{emb}} := \frac{1}{\|p\|}(I - ee^\top)J_\ell$ satisfies

$$J_{\text{emb}} \;=\; \frac{\partial \hat{p}}{\partial \delta}\Big|_{\delta=0}, \qquad e^\top J_{\text{emb}} = 0.$$

Proof. The Jacobian of the normalization $n(p) = p/\|p\|$ is $\partial n/\partial p = \frac{1}{\|p\|}(I - ee^\top)$ with $e = p/\|p\|$; composing with $J_\ell = \frac{1}{T}\partial p/\partial\delta$ by the chain rule gives $J_{\text{emb}} = \partial\hat{p}/\partial\delta$. For orthogonality, $e^\top(I - ee^\top) = e^\top - (e^\top e)e^\top = 0$ since $\|e\| = 1$. □

Radial scaling of p leaves $\hat{p}$ fixed, so the removed component $e^\top J_\ell$ — the first-order change in $\|p\|$ — is exactly the part the encoder discards; keeping it would inject a direction the downstream never sees.

Appendix B — Per-layer tables

Structural signatures and internal identity accuracy by residual depth, straight from the ledger.

Authors (MiniLM, 6 layers). Internal identity accuracy by layer: [5.2, 7.2, 8.8, 8.0, 7.2, 8.8]% (chance 1.8%).

layer	stable rank	eff dim	verbaliz.	autocorr
0	22.1	73.5	0.50	-0.004
1	31.4	88.8	0.51	+0.004
2	31.2	93.0	0.53	+0.048
3	37.2	104.6	0.50	+0.101
4	54.5	127.6	0.49	+0.116
5	106.9	205.2	0.61	+0.088

Coders (JinaBERT, 12 layers, 15 developers). Internal identity accuracy by layer: [45.2, 44.8, 48.0, 50.8, 40.0, 48.8, 48.0, 40.4, 42.0, 42.0, 42.4, 43.2]% (chance 6.7%).

layer	stable rank	eff dim	verbaliz.	autocorr
0	30.6	79.6	0.19	-0.033
1	36.7	95.2	0.20	-0.030
2	38.6	98.9	0.21	-0.001
3	35.6	100.7	0.21	+0.023
4	33.4	102.4	0.21	+0.032
5	20.3	91.2	0.21	+0.039
6	37.9	113.8	0.19	+0.062
7	54.3	126.4	0.20	+0.071
8	59.5	132.0	0.20	+0.086
9	61.7	139.0	0.22	+0.085
10	73.3	151.3	0.23	+0.088
11	79.0	185.0	0.22	+0.075

Decoder (GPT-2, 12 layers). Next-token logit-lens accuracy at the 13 residual points (input embedding, then 12 layers): [0.0, 2.6, 2.6, 3.2, 3.2, 4.7, 5.1, 7.2, 10.9, 13.0, 17.0, 21.3, 22.6]%. The four structural signatures below are the 12 layer-Jacobians (depths 0–11), so "the output" is the final residual (22.6%) for next-token accuracy and the last layer for the signatures.

layer	stable rank	eff dim	verbaliz.	autocorr
0	2.6	5.8	0.89	+0.007
1	5.0	16.2	1.81	+0.050
2	6.3	25.8	2.44	+0.031
3	6.4	30.3	2.08	+0.047
4	6.6	32.1	2.15	+0.044
5	6.7	33.5	1.94	+0.052
6	6.5	31.1	1.90	+0.058
7	6.5	29.9	1.57	+0.099
8	6.7	32.5	1.26	+0.123
9	5.2	23.7	1.39	+0.145
10	3.8	13.8	1.10	+0.161
11	2.0	4.1	0.89	+0.110

Appendix C — Axis definitions, rosters, and OOD probe sets

Prose style axes (`author_axes`): `male`↔`female` (gender difference-of-means); and the top-2 one-vs-rest classes of the `educated` and `raised` fields: `educated=England`↔`rest`, `educated=US`↔`rest`, `raised=England`↔`rest`, `raised=US`↔`rest`. Code style axes (`dataset_axes`): `Systems`↔`rest`, `Scripting`↔`rest` (the `paradigm` trait). Each axis is the L2-normalized difference of class-mean reference embeddings, a unit Fisher direction, used for both readout and steering.

Rosters. 55 public-domain prose authors (`corpus/authors.toml`, labelled with gender / birth / raised / educated / college) and 15 open-source developers (`corpus/coders.toml`, git-attributed and name-scrubbed, labelled with a `paradigm` = Systems/Scripting trait).

OOD probes for the fingerprint detector (Appendix E). Prose (bar 0.30): four held-out author samples plus `source code`, `modern chat`, `biology abstract`, `legalese`. Code (bar 0.51): four held-out coder samples plus `Victorian prose`, `modern chat`, `news headline`, `recipe`.

Appendix D — Audit ledger

Every number above is generated by `jlens/paper/build_ledger.py` into `jlens/paper/ledger.json` (value → source asset + JSON path) and cross-checked by the figure pipeline. Method claims resolve via `jlens/paper/method_map.md`.

Appendix E — Fingerprint-presence detector and AI-model fingerprinting

Two results that use the same identity geometry but sit off the paper's structural thesis; we record them here for completeness.

Fingerprint presence: known identity vs. "blank space". A calibrated nearest-centroid detector separates known identities from out-of-distribution text, asking a question the reference paper does not: is a given person's fingerprint present in the embedding at all? Prose (bar 0.30): known probes 0.62–0.69, out-of-distribution text (code, chat, biology, legalese) 0.10–0.25. Code is tighter (bar 0.51): known 0.54–0.70, OOD 0.17–0.49; "modern chat" sits just under the bar.

Figure 9: Is a person's fingerprint present in the embedding? Known identities vs. out-of-distribution "blank space".

Fingerprinting the AI models: the task dominates. If humans have fingerprints, do the *models* people code with? Here we have ground truth: we generate the code, so we know which model wrote it. We prompted four frontier models (`claude-opus-4.8`, `gpt-5.6-terra`, `gemini-3.5-flash`, `deepseek-chat`) across 40 coding tasks (24 canonical + 16 open-ended) and embedded the output in the same JinaBERT space as the humans (149 of the 160 model×task cells produced non-empty code).

Same-task, different-model code is alike (cosine 0.71), which *looks* like the frontier models have converged to one style, but a control refutes it: same-model / different-task similarity is only 0.20, so it is the task, not the model, that drives the embedding. Controlling for the task, leave-one-task-out nearest-model-centroid still recovers which model wrote unseen code at 42% ([34.6, 50.3] Wilson, n=149) against a 25% chance baseline, significantly above a within-task label-shuffling null (null mean 24%, p<0.001 over 1000 permutations). The

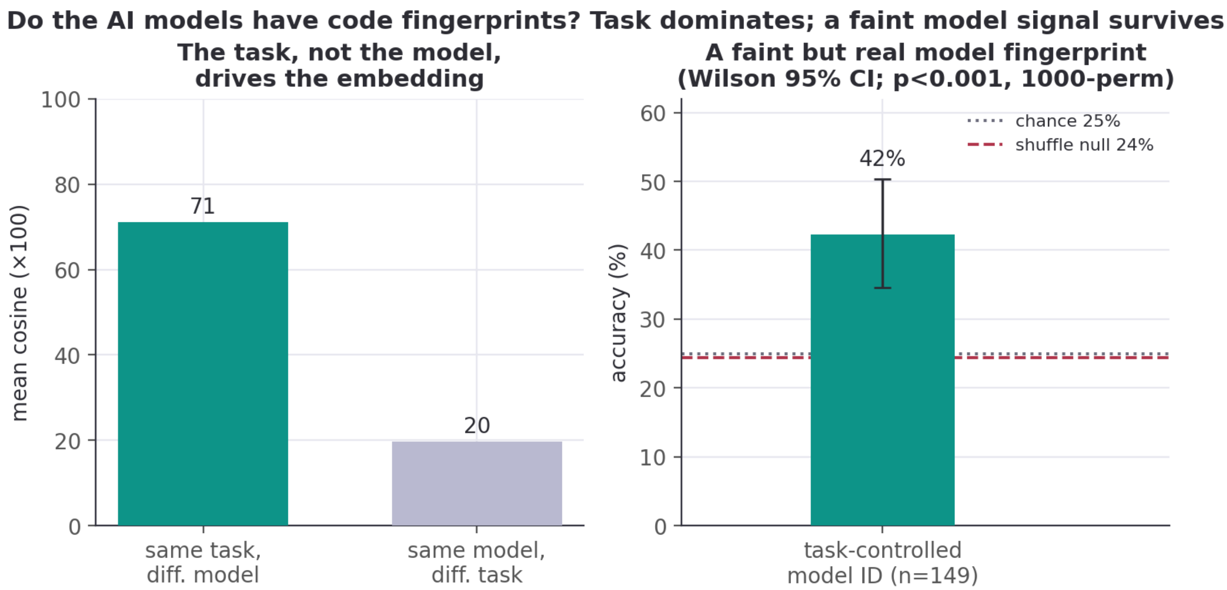

Figure 10: Task vs. model: on the same task, different models write nearly the same code (0.71); the same model across different tasks is far less alike (0.20). Controlling for the task, model identity is recoverable at 42% vs 25% chance.

models carry a detectable but faint style, largely masked by what the code *does*. This is the content–style control for the code side (§7); we make no convergence claim and no individual-usage claim.

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